

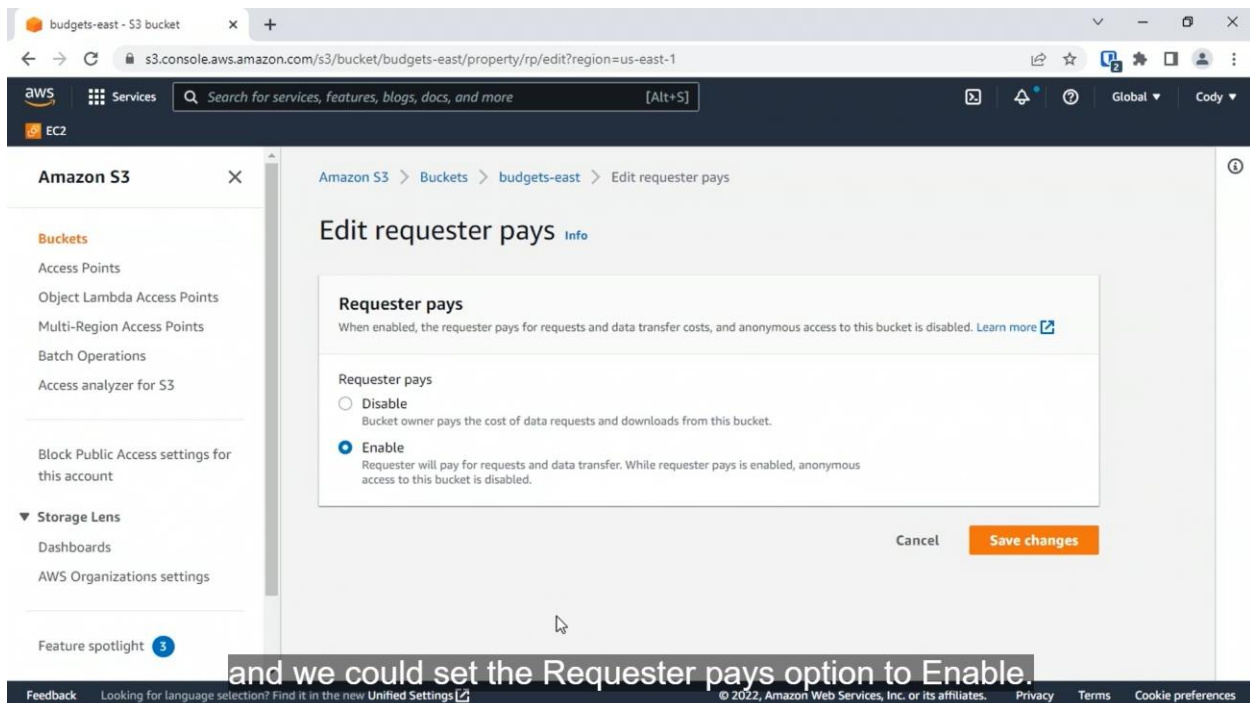
3.2 Cost Management

In this course, I will:

- Cover when to use various cost savings strategies and work with EC2 instance reservations, as well as spot instances
- Tag resources for cost allocation purposes using the console, the CLI, and PowerShell
- Analyze AWS charges using the Cost Explorer tool
- Create cost and usage budgets in order to track monthly expenditures and to be alerted when a certain threshold cost is reached

AWS Cost Management Strategies

Enabling S3 Requests and Transfers



Enable the requester pays option in buckets

```
AWS CloudShell Actions ⓘ  
us-east-1  
[cloudshell-user@ip-10-0-7-111 ~]$ aws s3api get-object help
```

Reducing EC2 Instance Costs with Spot Instances

Configuring Reserved Instances

Tagging AWS Resources with the CLI

```
AWS CloudShell Actions ⓘ  
us-east-1  
[cloudshell-user@ip-10-0-26-133 ~]$ aws ec2 run-instances --tag-specification 'ResourceType=instance,Tags=[{Key=Name,Value=Ubuntu1}]','ResourceType=volume,Tags=[{Key=CostCenter,Value=Toronto1}]' --image-id ami-052efd3df9dad4825
```

Add key value

```
AWS CloudShell Actions ⓘ  
us-east-1  
[cloudshell-user@ip-10-0-26-133 ~]$ aws ec2 create-tags --resources i-06900c4588a3fdcf5 --tags Key=Project,Value=XYZ
```

Route table tags

```
us-east-1  
[cloudshell-user@ip-10-0-26-133 ~]$ aws ec2 create-tags --resources i-06900c4588a3fdcf5 --tags Key=Project,Value=XYZ  
[cloudshell-user@ip-10-0-26-133 ~]$ aws ec2 create-tags --resources rtb-0a488ce9d916a616b --tags Key=Project,Value=XYZ  
[cloudshell-user@ip-10-0-26-133 ~]$
```

Using PowerShell to Tag AWS Resources

```
Administrator: Windows PowerShell  
PS C:\> get-command *tag*
```

Create tag, get tag

```
Administrator: Windows PowerShell
PS C:\> get-awscredential -listprofiledetail

ProfileName StoreTypeName      ProfileLocation
-----
admin1      NetSDKCredentialsFile
default     NetSDKCredentialsFile

PS C:\> initialize-awsdefaultconfiguration -profilename admin1
PS C:\> New-Ec2Tag -resourceid i-06900c4588a3fdcf5 -tag @{Key="CostCenter";Value="Toronto1"}
PS C:\> Get-Ec2Tag -region us-east-1 -filter @{Name="CostCenter";Value="Toronto1"}
Get-Ec2Tag : The filter 'CostCenter' is invalid
At line:1 char:1
+ Get-Ec2Tag -region us-east-1 -filter @{Name="CostCenter";Value="Toron ...
+ ~~~~~
+ CategoryInfo          : InvalidOperation: (Amazon.PowerShe...GetEC2TagCmdlet:GetEC2TagCmdlet) [Get-Ec2Tag]
+ FullyQualifiedErrorId : Amazon.EC2.AmazonEC2Exception,Amazon.PowerShell.Cmdlets.EC2.GetEC2TagCmdlet

PS C:\> Get-Ec2Tag -region us-east-1 -filter @{Name="tag:CostCenter";Value="Toronto1"}

Key          ResourceId      ResourceType Value
---
CostCenter   i-06900c4588a3fdcf5 instance      Toronto1
CostCenter   vol-0c8a10ebe7ced4808 volume        Toronto1

PS C:\> The tag itself is called CostCenter.
```

Analyzing Costs with AWS Cost Explorer Tool

Creating an AWS Cost Budget

1. Which CLI syntax can be used to tag as resources?
 - aws ec2 create-tags
2. You would like a list of resources tagged with a specific Project in a specific region. Which PowerShell cmdlet should you use?
 - Get-Ec2Tag

Remaining 7 questions same as 3.1